

4C

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1 Load in Data

Conversion of .h5afd file to csv can be found in 3a_3b_4c.py

```
counts <- read.csv('C:/Users/Trish/Documents/CMU/Comp Genome/HW1/counts_matrix.csv', row.names = 1, check.names = FALSE)
counts <- as.matrix(counts)

cat("Loaded:", nrow(counts), "genes x", ncol(counts), "cells\n")
```

```
## Loaded: 1531 genes x 13158 cells
```

```
# Create Seurat object
seurat_obj <- CreateSeuratObject(counts = counts, project = "intestine")
```

```
## Warning: Data is of class matrix. Coercing to dgCMatix.
```

```
# Compute library sizes
library_sizes <- colSums(counts)
seurat_obj@meta.data$library_size <- library_sizes
seurat_obj@meta.data$log_library_size <- log10(library_sizes)

# Verify it worked
cat("Library size added:", "library_size" %in% colnames(seurat_obj@meta.data), "\n")
```

```
## Library size added: TRUE
```

2 Apply standard log-normalization to the count matrix

```
seurat_obj <- NormalizeData(seurat_obj,
                           normalization.method = "LogNormalize",
                           scale.factor = 10000)
```

```
## Normalizing layer: counts
```

```
log_norm_data <- as.matrix(GetAssayData(seurat_obj, layer = "data"))
```

3 Apply scTransform()

```
seurat_obj <- SCTransform(seurat_obj, return.only.var.genes = FALSE)

## Running SCTransform on assay: RNA

## Running SCTransform on layer: counts

## vst.flavor='v2' set. Using model with fixed slope and excluding poisson genes.

## Variance stabilizing transformation of count matrix of size 1395 by 13158

## Model formula is y ~ log_umi

## Get Negative Binomial regression parameters per gene

## Using 1244 genes, 5000 cells

## There are 6 estimated thetas smaller than 1e-07 - will be set to 1e-07

## Second step: Get residuals using fitted parameters for 1395 genes

## Computing corrected count matrix for 1395 genes

## Calculating gene attributes

## Wall clock passed: Time difference of 30.03314 secs

## Determine variable features

## Centering data matrix

## Getting residuals for block 1(of 3) for counts dataset

## Getting residuals for block 2(of 3) for counts dataset

## Getting residuals for block 3(of 3) for counts dataset

## Centering data matrix

## Finished calculating residuals for counts

## Set default assay to SCT
```

```
sct_data <- as.matrix(GetAssayData(seurat_obj, assay = "SCT", layer = "scale.data"))
```

4 Calculate Gene-Wise Correlation

```
common_genes <- intersect(rownames(log_norm_data), rownames(sct_data))

log_cors <- sapply(common_genes, function(gene) {
  cor(log_norm_data[gene, ], library_sizes, use = "complete.obs")
})

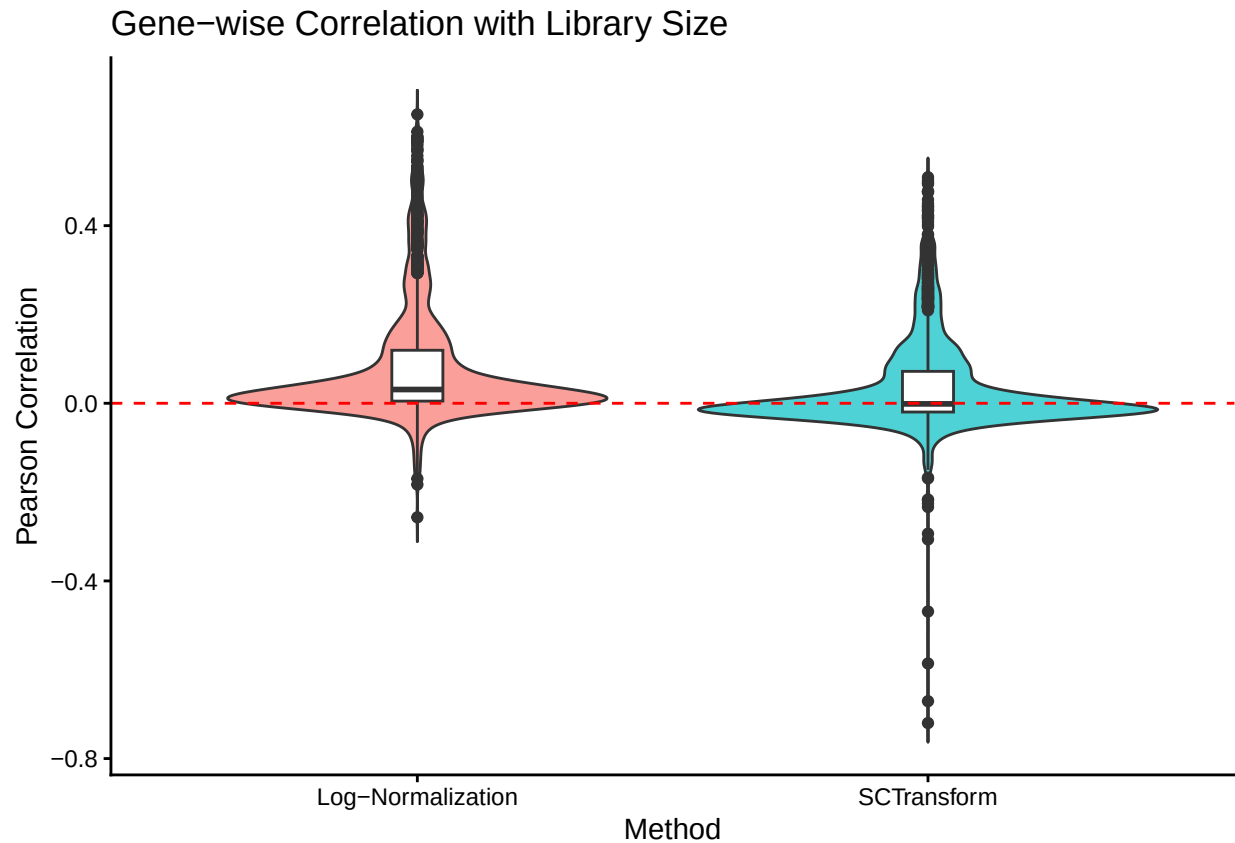
sct_cors <- sapply(common_genes, function(gene) {
  cor(sct_data[gene, ], library_sizes, use = "complete.obs")
})
```

5 Generate plots

```
cor_df <- data.frame(
  Method = rep(c("Log-Normalization", "SCTransform"),
    each = length(log_cors)),
  Correlation = c(log_cors, sct_cors)
)

p3 <- ggplot(cor_df, aes(x = Method, y = Correlation, fill = Method)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_boxplot(width = 0.1, fill = "white") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Gene-wise Correlation with Library Size",
    y = "Pearson Correlation") +
  theme_classic() +
  theme(legend.position = "none") +
  scale_fill_manual(values = c("#F8766D", "#00BFC4"))

p3
```



```
p4 <- ggplot(cor_df, aes(x = Correlation, fill = Method)) +
  geom_histogram(bins = 50, alpha = 0.8) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  facet_wrap(~Method, ncol = 1) +
  labs(title = "Distribution of Gene-Library Size Correlations",
       x = "Correlation") +
  theme_classic() +
  theme(legend.position = "none") +
  scale_fill_manual(values = c("#F8766D", "#00BFC4"))

ggsave("correlation_histogram.png", p4, width = 10, height = 8)
p4
```

Distribution of Gene–Library Size Correlations

